Unofficial English translation, made for demonstration purposes. Layout simplified; this is not the original document. Original: tijhuisingenieurs.nl/files/ti20332-pres-04-technieken-bij-de-pilots-2-v3.pdf

4.0

existing legakker

North side South side

North side South side

4.0 existing legakker

North side South side

4.0 existing legakker

4.0

North side South side

existing legakker

North side South side

4.0 existing legakker

4.0 submerged legakker

North side South side

Techniques used in the pilots

Pilot 4: Protective bank along a busy navigation route

A nature-friendly bank along a recognised navigation route in the northern Kievitsbuurt. The legakker will be fitted with a reed bank on one side (the south).

Method

We prune the existing legakker and remove the stumps.

We install bank protection (two techniques).

The bank protection is lined with maize cloth or jute cloth.

Dredged material is introduced.

Once the dredged material has consolidated, we plant reed.

After 4–5 years, we remove the bank protection down to below the waterline and finish the slope.

We use two bank protection techniques:

1. We place a row of piles with planks and a waling, between which we introduce dredged material. After a few years, we remove the top plank between the piles.

2. The second technique is similar to the first. We place a continuous row of piles with a waling. The waling remains above water.

Pilot 5: Reed growth in combination with dredged material

Legakker in the northern Kievitsbuurt. Within this pilot, different quantities of dredged material are used.

Method

We prune the existing legakker and remove the stumps.

We place a continuous row of piles.

The row of piles is lined with maize cloth or jute cloth.

Different quantities of dredged material are introduced.

Once the dredged material has consolidated, we plant reed.

After 4–5 years, we remove the bank protection down to below the waterline and finish the slope.

We test three different depths for reed growth:

1. We introduce dredged material until the water depth within the piles is 0.35 m.

2. Between the rows of piles, we introduce (dead) (willow) branches. We introduce dredged material until the water depth within the piles is 0.35 m.

3. We introduce dredged material at two different levels. Between the rows of piles, we place anchored straw bales.

Pilot 6: Restoration of legakker and construction of protective foreshore

Construction of a legakker and a foreshore on the south side with a slope of 1:5 in the northern Kievitsbuurt.

Method

At 4 metres from the legakker, we place a continuous row of piles.

The row of piles is lined with maize cloth or jute cloth.

We introduce dredged material.

Once the dredged material has consolidated, we form a slope (incline) of 1:5, on which reed is planted.

After 4–5 years, we remove the bank protection down to below the waterline and finish the slope. We push the row of piles away.

North side: new bank protection is placed in front of the existing bank protection, up to above water level.

Technique 1 Technique 2

Technique 1 Technique 2 Technique 3

Photo of location

Photo of location

Photo of location